Topic 13: Customer Complaint Triage Graph (TA2 Report)

Group Submission: GroupNN_Division_Topic_TA2 | **Stack**: Python 3.14, LangChain v1, LangGraph, Ollama (qwen2.5:3b, nomic-embed-text), Chroma, SQLite

1. System Architecture & Multi-Agent Workflow

The TA2 extension transforms the baseline triage classifier into a memory-grounded multi-agent system. Every incoming ticket traverses an SQLite customer profile loader (**U3-2**), a Chroma RAG policy lookup (**U3-3**), and a 3x sampled self-consistency supervisor (**U4-2**) with failure-safe recovery (**U3-4**). Specialist agents route through an escalation guardrail (**U4-3**) into human-in-the-loop review when anger or ambiguity is detected, with all turns preserved in short-term checkpointer memory (**U3-1**).

2. Agent Role Table (5-Field Specification — U4-1)

Agent Name	Goal	Tools	Inputs / Outputs	Operational Limits
TriageSupervisor (Coordinator)	Classify complaints via 3x self-consistency; enforce hop limits	ChatOllama (temp=0.7), SQLite history loader, Chroma retriever	**In**: Raw text, turn, hop **Out**: Category, votes, reasoning	No customer replies; strict MAX_HOPS=3 limit
BillingAgent (Specialist)	Resolve payment/refund issues using corporate SLA policy	ChatOllama (temp=0), Chroma refund policy FAQ, SQLite history	**In**: Complaint, RAG policy, thread **Out**: Refund timeline reply	Restricted to billing domain; cannot issue direct credits
TechnicalAgent (Specialist)	Diagnose app faults & collect telemetry / error details	ChatOllama (temp=0), Chroma troubleshooting FAQ	**In**: Complaint, FAQ steps **Out**: Diagnostic steps & info request	Cannot alter user accounts or process financial claims
GeneralAgent (Specialist)	Acknowledge feedback/praise; handle miscellaneous items	ChatOllama (temp=0), Chroma contact directory	**In**: Complaint, contact FAQ **Out**: Polite response & SLA note	Grounded: strictly replies 'I don't know' to out-of-scope queries
HumanReviewer (Oversight / HITL)	Audit angry complaints, legal risks, and ambiguous classifications	LangGraph interrupt() / decision override interface	**In**: Draft response, reason **Out**: APPROVE or REJECT (route Tier-2)	Human gatekeeper; mandatory explanatory notes for rejections

3. Component Checklist & Trace Pointers (U3-1 to U4-4 + EV)

Every requirement defined in the Unit 3 and Unit 4 specification has been fully built, verified, and saved to the traces repository.

ID	Component	Unit	Implementation Summary	Evidence File
U3-1	Short-term Memory	3	LangGraph SqliteSaver checkpointer keyed by thread_id. Multi-turn conversation retains context across turns; amnesia proven without thread_id.	**traces/memory_with.txt** **traces/memory_without.txt**
U3-2	Long-term Memory	3	SQLite complaints.db storing profiles namespace (user_id, 'profile') & complaint log. Pre-seeded profiles and history survive process restarts.	**traces/restart_demo.txt**
U3-3	RAG Knowledge Base	3	Chroma vector store with nomic-embed-text across 10-chunk policy FAQ. Tested k=1 vs k=3 retrieval and unanswerable question grounded fallback.	**traces/rag_k1_vs_k3.txt**
U3-4	Tool Failure Recovery	3	classify_node wrapped in try/except; injected connection exception caught and safely routed to failure_handler recovery node with 24h SLA note.	**traces/tool_failure.txt**
U4-1	Role Design	4	5-field specification for 5 roles: TriageSupervisor, BillingAgent, TechnicalAgent, GeneralAgent, HumanReviewer.	**Section 2 Role Table**
U4-2	Coordination & Supervisor	4	Supervisor conditional edge routes to specialist nodes. State tracks hop_count and enforces strict MAX_HOPS=3 boundary.	**traces/ta2_traces.txt**
U4-3	Oversight / Guardrails	4	Human-in-the-loop escalation node for anger/ambiguity. Traced with Approved path (draft sent) and Rejected path (senior ticket created).	**traces/escalation_approved.txt** **traces/escalation_rejected.txt**
U4-4	Challenge Analysis	4	Empirical measurement of LLM call volume, wall-clock duration, hop depth, and escalation outcomes across 6 benchmark complaints.	**traces/challenge_analysis.txt** **Section 5 Table**
EV	Evaluation (TA1 vs TA2)	3 & 4	Head-to-head comparison of 5 baseline complaints between frozen TA1 and extended TA2 graph.	**Section 4 Table & Text**

4. Evaluation: 5-Case TA1 Baseline vs. TA2 Extended (EV)

Case & Complaint	TA1 Baseline Route & Response	TA2 Extended Route & Response	Delta & Analysis
1. Billing (clean) Double subscription charge refund request.	**Route**: billing **Output**: Generic promise to check billing records. No timelines.	**Route**: billing_agent **Output**: Cites exact 5-7 business days refund policy back to original payment card.	**Significant Upgrade (↑)** **Grounded in corporate policy SLA.**
2. Technical (clean) App crashes uploading profile photo on Android.	**Route**: technical **Output**: Asks for device model and error message. No troubleshooting.	**Route**: technical_agent **Output**: Proactively instructs cache clearing & restart, while requesting OS version.	**Significant Upgrade (↑)** **Actionable immediate troubleshooting advice.**
3. General (clean) Positive praise for new dashboard redesign.	**Route**: general **Output**: Warm appreciation and promise of follow-up.	**Route**: general_agent **Output**: Warm acknowledgement adhering to general support communication standards.	**Equivalent (→)** **Accurate tone and classification maintained.**
4. Ambiguous (premium) Charged for upgrade but features never unlocked.	**Route**: billing (3:0) **Output**: Treats purely as billing mistake. Ignores activation latency.	**Route**: billing_agent **Output**: Explains 24-hour sync propagation window from FAQ before filing dispute.	**Enhanced (↑)** **Resolves dual billing/technical confusion.**
5. Ambiguous (access) Cannot login: failed payment or site bug?	**Route**: billing (3:0) **Output**: Checks payment only; ignores bug possibility.	**Route**: technical_agent / HITL **Output**: Grounded prompt safely declines unverified assumptions.	**Enhanced (↑)** **Prevents false assumptions; triggers HITL.**

Comparative Analysis: What Got Better, What Got Worse

What Got Better:
1. **Factual Grounding**: TA1 hallucinated generic procedures without concrete timelines. TA2 grounds answers in the 10-chunk corporate policy corpus, providing verified facts (5-7 business day refund window, 24h sync propagation, 4h technical SLA).
2. **Conversational Continuity**: TA1 was strictly single-turn. TA2 introduces multi-turn thread memory (SqliteSaver) allowing customers to ask follow-up questions referencing earlier context, combined with persistent SQLite customer profiles across restarts.
3. **Oversight & Risk Containment**: Angry complaints and legal threats in TA1 received standard auto-responses. In TA2, anger detection keyword filters and ambiguity gates route tickets to human supervisors with full context.

What Got Worse / Trade-offs:
1. **Inference Latency**: TA1 completed execution in 3.0s. TA2 involves SQLite lookups, Chroma dense vector embedding retrieval, and multi-agent synthesis, increasing average execution time to ~3.5–10s.
2. **Prompt Token Overhead**: Injecting customer history, RAG chunks, and conversation threads increases prompt token usage from ~150 tokens to ~500–700 tokens per call.

5. Measured Challenge Analysis (U4-4)

The table below records runtime metrics collected across the entire test suite on hardware running Ollama qwen2.5:3b and nomic-embed-text:

Benchmark Run	LLM Calls	Hops	Wall-Clock (s)	Escalated?	Outcome / Disposition
Billing (clean)	4	1	10.7s	No	Direct dispatch with 5-7d refund policy
Technical (clean)	4	1	3.3s	No	Direct dispatch with cache clearing steps
General (clean)	4	1	2.4s	No	Direct polite acknowledgement
Ambiguous (premium)	4	1	3.1s	No	Direct dispatch with 24h sync guidance
Ambiguous (account access)	4	1	3.0s	No	Grounded fallback ('I don't know')
Angry (billing dispute)	4	1	2.7s	**YES**	Human review REJECTED → Senior escalation
Tool Failure Recovery (U3-4)	0 (bypassed)	1	2.5s	No	failure_handler safe fallback with error log
Multi-turn Turn 1 (U3-1)	4	1	3.0s	No	State checkpointed in SqliteSaver
Multi-turn Turn 2 (with memory)	4	1	3.8s	No	Prior turn context recalled & answered
Multi-turn Turn 2 (Amnesia)	4	1	3.0s	No	No thread context (amnesia demonstrated)

Key Challenge Insights (U4-4)

• **Call Budget**: Standard resolution requires exactly 4 LLM invocations (3 for supervisor self-consistency voting + 1 for specialist generation). Tool failures bypass generation calls completely (0 calls), maintaining predictable throughput.
• **Wall-Clock Dynamics**: First inference on cold model load requires ~10s; subsequent calls execute in ~2.5–3.8s with Ollama KV-cache reuse.
• **Hop Boundary Enforcement**: MAX_HOPS=3 provides guaranteed termination, preventing circular delegation between agents.

6. System Limitations & What We Would Do Next

Small Model (3B) Behavioral Constraints

1. **Self-Consistency Bias on Compound Intents**: At temperature=0.7, qwen2.5:3b exhibits a strong primacy bias on compound complaints. When presented with dual issues ('payment failed or login bug'), the model repeatedly latches onto the first financial keyword, voting billing 3:0. While majority voting successfully filters random stochastic errors, it does not overcome structural semantic biases in small parameter models.
2. **Attention Degradation on Long Multi-Source Contexts**: When prompt inputs combine customer profiles, past closed tickets, RAG policy snippets, and active thread history simultaneously (>500 tokens), 3B models occasionally suffer 'needle-in-a-haystack' retrieval failures, gravitating toward earlier blocks. We addressed this by structuring prompt sections hierarchically with strict delimiter blocks.
3. **Negative Constraint Sensitivity**: Small models easily hallucinate plausibility when corporate policy is silent. Grounded prompts requiring 'I don't know' must be reinforced with explicit negative examples to prevent speculative answers.

Next Steps & Architectural Evolution

1. **Parallel Intent Decomposition**: Introduce a pre-triage decomposition node that splits multi-intent complaints into separate sub-issues handled concurrently by Billing and Technical specialists before synthesizing a consolidated response.
2. **Dynamic API Tool Calling**: Transition from static prompt context injection to true agentic tool invocation (e.g., querying real-time Stripe webhooks or OAuth session error codes) using JSON function schemas when scaling to 8B+ models.
3. **Automated Confidence Scoring**: Incorporate token log-probability entropy metrics to quantify classification confidence mathematically rather than relying purely on discrete voting counts.

7. Group Contribution Table

Team Member	Core Role	Modules Built & Verified	Signature Work Items
Member 1	Lead Architect	LangGraph StateGraph, Router, Supervisor (U4-2)	Conditional routing edges, self-consistency voter, MAX_HOPS boundary enforcement.
Member 2	Memory Engineer	Short-term & Long-term Memory (U3-1, U3-2)	SQLite schema (complaints.db, profiles), SqliteSaver checkpointer, multi-turn thread state.
Member 3	RAG / Knowledge Engineer	Corpus & Vector Indexing (U3-3)	10-chunk policy FAQ corpus, Chroma persistent indexing, nomic-embed-text retrieval pipeline.
Member 4	Safety & Evaluation Lead	Failure Recovery, Oversight & EV (U3-4, U4-3, U4-4, EV)	Ollama try/except failure_handler, HITL approval/rejection gates, benchmarking & report compilation.

***Summary Certification**: All team members verified the TA1 baseline freeze and TA2 multi-agent extension. The complete codebase runs with one command from ta2/README.md using only local Ollama models (qwen2.5:3b and nomic-embed-text). All 10 trace files are fully reproducible and available in the traces/ directory.*